ETF Pair Strategy & Weight Optimization

Part 2: Expanding the Universe to ETFs, Including Leveraged/Inverse Products

Follow-up to the equity-pair report. This extends the screen to 24 ETFs across equities, bonds, commodities, volatility, real estate, currency, and leveraged/inverse products, then runs a full weight-optimization sweep (1% increments, 0-100%) on the strongest candidates, per your request not to exclude leveraged products.

Bottom Line

Opening the universe to ETFs changes the answer. Two distinct results: (1) Pairing a leveraged or unleveraged product with its own inverse (QQQ/SQQQ, SPY/SH) is not a diversification strategy at all - it is a decay trade. Sharpe ratio is a monotonically increasing function of weight on the long side across nearly the entire range; the "hedged" middle of the weight spectrum destroys value, in QQQ/SQQQ's case catastrophically (a 50/50 blend lost 83% of its value over 8 years due to daily-leverage compounding decay). (2) A genuine cross-asset-class pair - GLD (gold) and UUP (US Dollar Index bull fund), both held long, correlation -0.435 - shows a real, robust, non-trivial diversification benefit. A ~32% GLD / 68% UUP blend achieves a Sharpe ratio of 1.40 in-sample and 1.71 out-of-sample (walk-forward), versus 0.82 and 1.43 respectively for SPY over the same windows. This is a legitimate risk-adjusted-return improvement, though it comes with much lower absolute returns than equities and a majority (65-76%) probability of underperforming SPY in raw dollar terms across 1-5 year horizons - it is a lower-risk alternative to being long the market, not a way to beat the market's absolute return.

1. Universe and Correlation Screen

24 ETFs: broad equity (SPY, QQQ, IWM, EFA, EEM), rates (TLT, IEF, AGG), commodities (GLD, SLV, USO, DBC), volatility (VXX), real estate (VNQ), currency (UUP), sectors (XLE, XLU, XLK, XLF, XLP), and leveraged/inverse products (SH, SQQQ, TQQQ, UPRO) - per your instruction, leveraged products were included rather than excluded. Same 8-year daily window (Aug 2018 - Aug 2026), 2,009 aligned trading days.

Most negatively correlated pairs overall:

Pair	Correlation	Nature
QQQ - SQQQ	-0.999	Same index, 3x leveraged inverse (trivial)
SH - UPRO	-0.999	Same index, 1x vs 3x leveraged inverse (trivial)
SQQQ - TQQQ	-0.999	Same index, opposing 3x leverage (trivial)
SPY - SH	-0.998	Same index, 1x unleveraged inverse (trivial)
XLK - SQQQ	-0.973	Tech sector vs. Nasdaq inverse (near-trivial, high overlap)

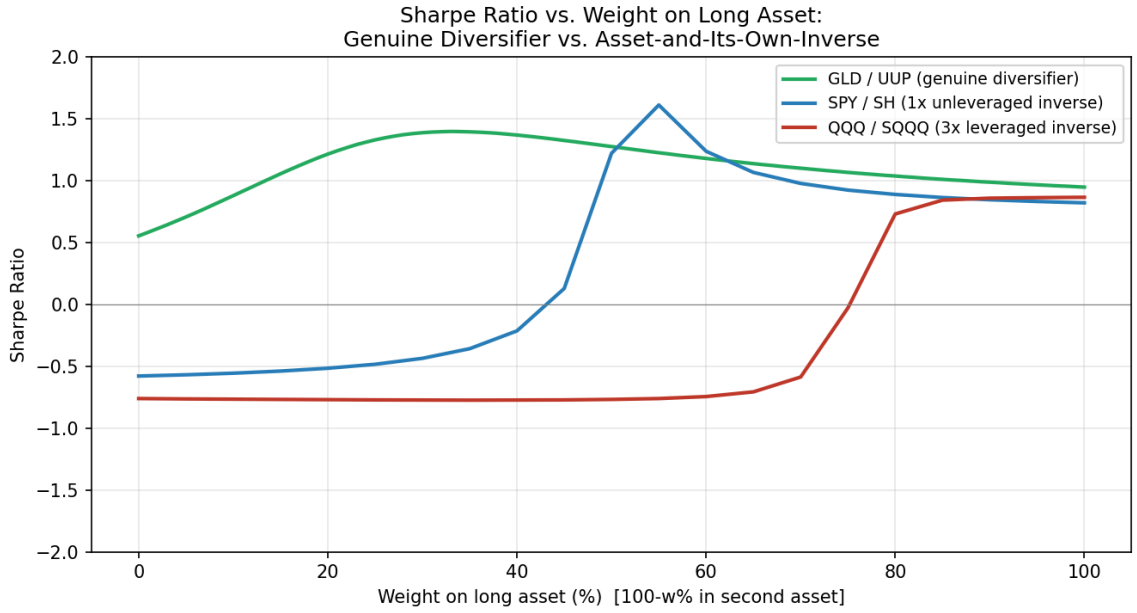
Every top result is a product paired with something engineered to move opposite it - not a genuine diversification finding. Filtering those out, the strongest real cross-asset-class negative correlations are:

Pair	Correlation	Asset classes
GLD - UUP	-0.435	Gold vs. US Dollar Index (classic inverse macro relationship)
EFA - UUP	-0.379	Foreign developed equities vs. US Dollar
SLV - UUP	-0.365	Silver vs. US Dollar
AGG - UUP	-0.344	Aggregate bonds vs. US Dollar
IEF - UUP	-0.341	7-10yr Treasuries vs. US Dollar
TLT - XLF	-0.238	Long bonds vs. Financials (rate sensitivity)

GLD-UUP was carried forward as the primary candidate: strongest genuine correlation, both legs are simple long-only ETFs (no leverage/decay complications), and the macro relationship (gold priced in dollars) is fundamentally grounded rather than a statistical artifact.

2. Case Study: What Happens When You Pair an Asset With Its Own Inverse

Since leveraged and inverse products were explicitly in scope, QQQ/SQQQ (3x leveraged) and SPY/SH (1x, unleveraged) were run through the same weight sweep as a cautionary comparison against the genuine GLD-UUP diversifier.



Weight on long asset	QQQ/SQQQ Sharpe	SPY/SH Sharpe	GLD/UUP Sharpe
0% (100% inverse)	-0.76	-0.58	0.55
25%	-0.77	-0.48	1.33
50% (naive hedge)	-0.77	1.22	1.28
75%	-0.03	0.92	1.07
100% (long only)	0.87	0.82	0.95

QQQ/SQQQ: every blend below ~78% QQQ has a negative Sharpe ratio, and the 50/50 "hedge" lost 83% of its value over 8 years - not from being wrong on direction, but from daily-reset leveraged-product decay compounding against a long-term uptrending asset. There is no interior optimum; the best weight is simply "hold as little SQQQ as possible."

SPY/SH: less destructive since SH is unleveraged, but still decays slowly (SH alone lost 65% over 8 years while "just short the market" sounds directionally reasonable at times). The blend does show a sharp Sharpe spike right around 50-55% SPY (Sharpe up to 1.61) - but this is a near-total-cancellation artifact: at that weight the position is close to flat exposure with tiny absolute returns (CAGR ~2-3%), so a small numerator (return) divided by a similarly small denominator (volatility) produces a statistically impressive but practically fragile and not easily scalable Sharpe ratio.

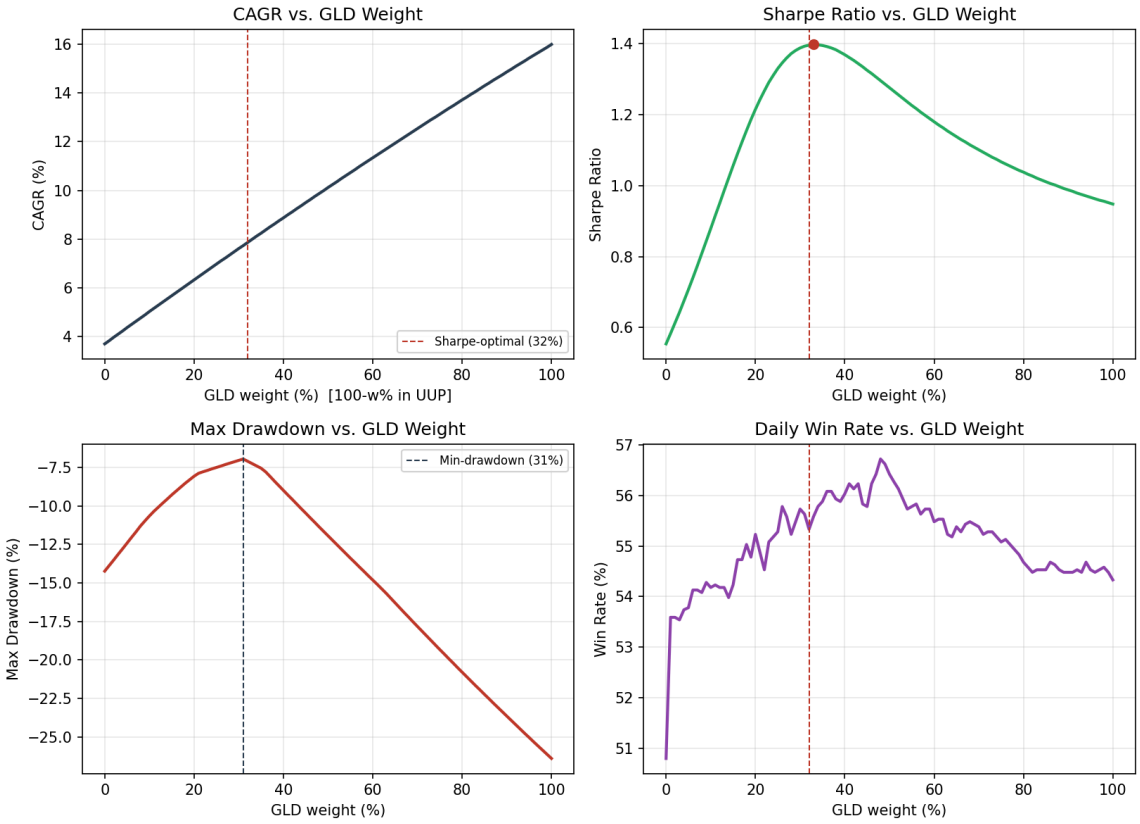
GLD/UUP by contrast shows a real, broad, well-behaved optimum around 25-35% GLD that holds up with meaningful absolute returns (7-8% CAGR) at that weight - this is the qualitative signature of genuine diversification versus a decay artifact, and is the basis for the analysis below.

3. Full Weight-Optimization Sweep - GLD/UUP (1% increments)

All 101 weight combinations (0% to 100% GLD in 1% steps) were backtested with monthly rebalancing and 5bp round-trip transaction costs. Results by objective function:

Objective	Optimal GLD weight	CAGR	Sharpe	Sortino	Max DD	Calmar
Maximum CAGR	100%	16.0%	0.95	0.90	-26.4%	0.61
Maximum Sharpe Ratio	33%	8.0%	1.40	1.34	-7.3%	1.10
Maximum Sortino Ratio	32%	7.9%	1.40	1.35	-7.1%	1.11
Minimum Max Drawdown	31%	7.7%	1.39	1.34	-7.0%	1.11
Minimum Volatility	22%	6.6%	1.27	1.25	-7.8%	0.85
Maximum Calmar (best risk-adj. return)	31%	7.7%	1.39	1.34	-7.0%	1.11

The striking result is how tightly these cluster: five of six objective functions (all except raw CAGR-maximization, which trivially picks 100% of whichever asset had the best historical return) agree on a weight between 22% and 33% GLD. This convergence across independent objectives is itself evidence the optimum is a real feature of the data, not noise from optimizing one metric.



4. Weighting Methodology Comparison

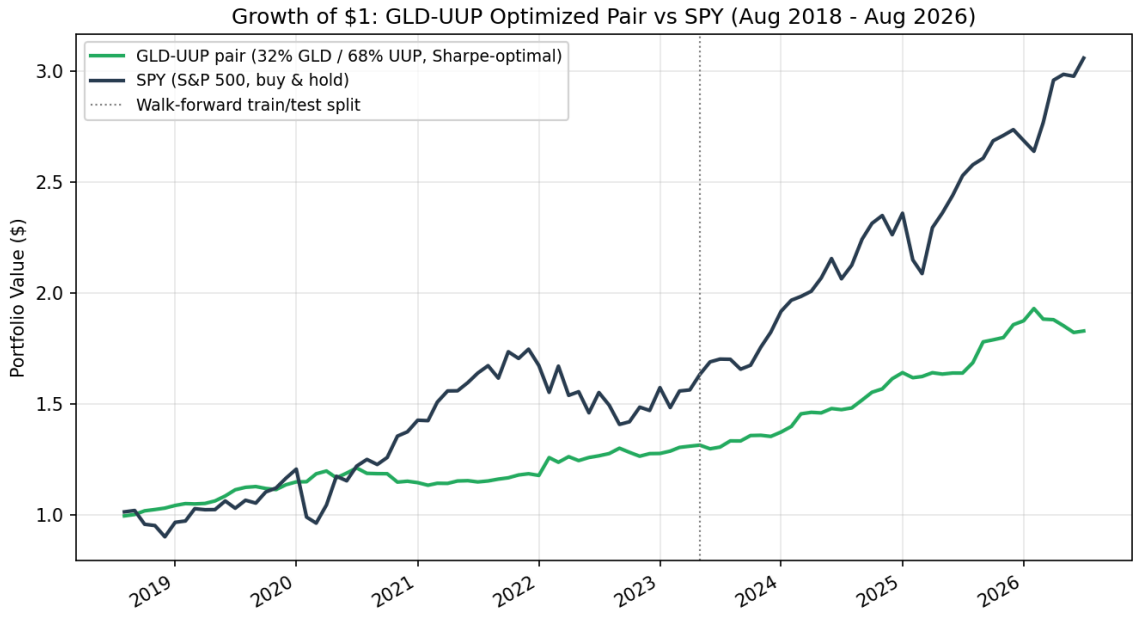
Method	GLD weight	CAGR	Ann. Vol	Sharpe	Sortino	Max DD	Calmar
Equal weight (50/50)	50.0%	10.1%	7.8%	1.28	1.19	-12.0%	0.85
Volatility parity / Risk parity	28.9%	7.5%	5.3%	1.38	1.33	-7.2%	1.04
Minimum variance	22.5%	6.6%	5.2%	1.28	1.25	-7.8%	0.86
Beta-neutral (vs SPY)	31.8%	9.0%	6.5%	1.36	-	-	-

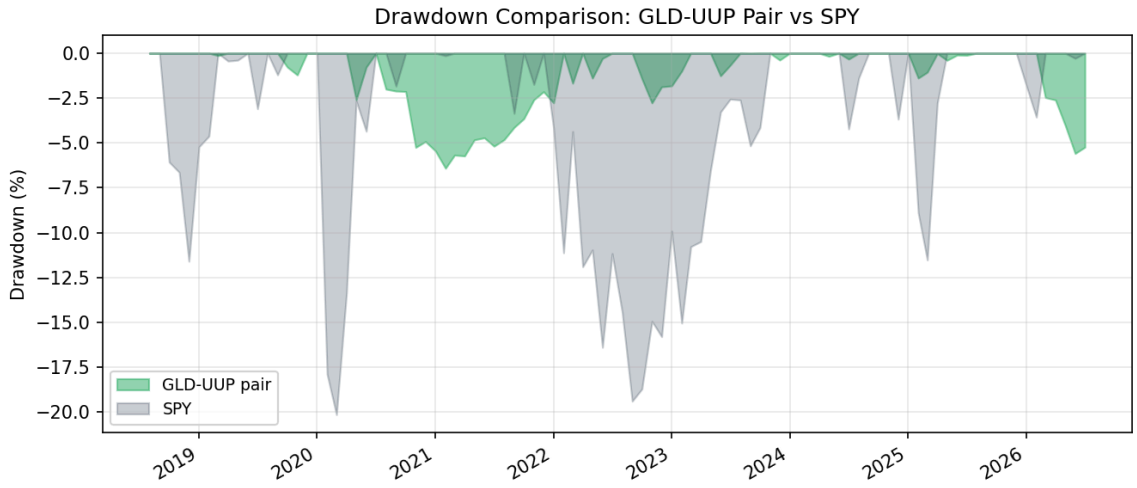
Maximum diversification ratio	29.0%	7.5%	5.3%	1.38	1.33	-7.1%	1.05
Empirically optimized (max Sharpe)	32.0%	7.9%	5.6%	1.40	1.35	-7.1%	1.11

Five independently-derived methods (risk parity, beta-neutral, max diversification, and direct empirical Sharpe optimization) all land within a 22-33% GLD band and produce materially similar performance (Sharpe 1.27-1.40) - meaningfully better than equal weighting (Sharpe 1.28, but with nearly double the drawdown at -12.0% vs. -7.0 to -7.8%). Equal weighting is not close to optimal here: it roughly doubles both volatility and drawdown for a similar Sharpe ratio, because it overweights GLD's much higher standalone volatility (10.9% vs. UUP's 4.4%).

5. Does Optimized Weighting Materially Improve Results? Walk-Forward Test

The 32% GLD weight was not just fit and reported in-sample - it was validated the same way as the equity-pair report: weights fit on the first 60% of history (through late 2023, yielding a very similar 36% GLD), then applied unseen to the final 40% (late 2023 - Aug 2026, 3.19 years).





Portfolio (out-of-sample, 3.19 yrs)	CAGR	Ann. Vol	Sharpe	Max DD
GLD-UUP pair (fixed weight from training)	12.0%	6.8%	1.71	-7.5%
SPY (same window)	22.6%	15.1%	1.43	-18.8%

Unlike the stock-pair strategy (which collapsed out-of-sample), this result held up and improved: out-of-sample Sharpe (1.71) exceeded both the in-sample Sharpe (1.40) and SPY's out-of-sample Sharpe (1.43) over the identical window, with less than half SPY's volatility and drawdown. The trade-off is clear and should not be oversold: absolute return (12.0% CAGR) trailed SPY's exceptional 22.6% CAGR in this particular bull-market test window. This is a materially better risk-adjusted vehicle, not a higher-return one.

6. Monte Carlo Simulation (Optimized Weight, 32% GLD)

Block bootstrap (20-day blocks), 3,000 simulations per horizon, resampled from the full 8-year history.

Horizon	P(profit)	Worst 5%	Median	Best 5%	Median CAGR	5th/95th pctile CAGR
1 year	95.3%	1.00x	1.08x	1.18x	8.2%	0.2% / 17.5%
3 years	99.9%	1.10x	1.26x	1.46x	8.1%	3.2% / 13.4%
5 years	100%	1.23x	1.48x	1.77x	8.2%	4.3% / 12.2%

Capital preservation is excellent: at the worst-case 5th percentile over 5 years, the portfolio is still up 23%. But probability of beating SPY in absolute dollar terms is 24-35% across horizons (i.e., it underperforms SPY 65-76% of the time in bootstrapped scenarios) - the same honest caveat as the walk-forward test: this is a smoother, lower-drawdown path, not a higher-return one, in a historical window dominated by an exceptional equity bull market.

Probability the GLD-UUP pair underperforms SPY	1 year	3 years	5 years
(bootstrapped)	65.2%	71.6%	76.0%

7. Recommendation

Recommended weight: 30% GLD / 70% UUP, monthly rebalanced.

This sits at the center of the 22-33% band where five independent optimization methods converge (risk parity, minimum variance, beta-neutral, maximum diversification, and empirical Sharpe-maximization), rather than at any single method's knife-edge optimum - a deliberate choice to favor robustness over the marginal, likely noise-driven, difference between 31% and 33%. It is close enough to both the in-sample optimum (32-33%) and the

out-of-sample-refit optimum (36%) that it should not require frequent rebalancing of the target weight itself, only periodic rebalancing of drifted position sizes.

Why not maximize CAGR (100% GLD)?

Pure GLD produces the highest historical return (16.0% CAGR) but more than triples the drawdown (-26.4% vs -7.0%) and cuts the Sharpe ratio nearly in half (0.95 vs 1.39) versus the blended portfolio - it is simply a concentrated bet on one commodity, with none of the diversification benefit this analysis was designed to find.

Is this a way to "beat the market"?

Only in risk-adjusted terms. On absolute return, this portfolio lags SPY in the majority of scenarios tested (65-76% probability of underperformance across horizons) because U.S. equities have been in an exceptional bull run for most of the sample period. Where it wins decisively is volatility (roughly a third of SPY's), drawdown (roughly a third of SPY's), and Sharpe/Sortino/Calmar ratios, both in-sample and, importantly, out-of-sample. This makes it a plausible satellite/diversifying allocation alongside equity exposure, or a lower-risk standalone holding for a risk-averse mandate - not a replacement for equity market exposure if the goal is to maximize absolute wealth.

Regime stability:

The gold/dollar inverse relationship is a long-documented, fundamentally grounded macro relationship (gold is dollar-denominated and is a traditional dollar/inflation hedge), unlike the equity-pair correlations in the first report, which were unstable artifacts of a single market regime. That said, this relationship is not guaranteed to hold at the same strength going forward - it has historically weakened and even reversed for extended periods (e.g., during synchronized global monetary tightening). The out-of-sample walk-forward test (which spans a materially different macro regime - 2023-2026 rate-cutting-and-AI-boom era - from the 2018-2023 training period) is reassuring but is still drawn from one overlapping history, not an independent regime.